CP-Free ODDM over General Doubly-Selective Fading Channels

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***Abstract*—This paper proposes a new orthogonal delay-Doppler division multiplexing (ODDM) system, which is designed to operate without the need of using a cyclic prefix (CP) in the transmitted signals. The CP-free ODDM is enabled by exploiting the unique structures of the ODDM prototype pulses, which are constructed through repetitions of finite-duration elementary pulses. The system eliminates the need of CP by designing a new receiving filter through temporally extending the transmission prototype pulse. The elementary pulse repetition at the transmitter along with the temporal filter extension at the receiver introduce a wrap-around effect in the equivalent channel spreading function in the dealy-Doppler (DD) domain. The wrap-around effect leads to a block-circulant-like structure of the DD-domain channel matrix that is the same as conventional ODDM systems with CP. Thus CP-free ODDM can employ the same receiver as conventional ODDM systems, yet it yields a higher energy efficiency as no energy needs to be allocated for CP symbols. The wrap-around effects are theoretically demonstrated by deriving the exact analytical expressions of the cross-ambiguity functions of practical transmission and receiving prototype pulses. In addition, theoretical analysis of the cross-ambiguity function verifies that the CP-free ODDM waveforms satisfy local bi-orthogonality in the delay and Doppler domains. This bi-orthogonality property along with the wrap-around effects of the channel spreading function enables the design of a low complexity iterative receiver, which performs interference cancellation and coherent matrix combining (CMC) in the delay domain and minimum mean squared error (MMSE) detection in the Doppler domain. The proposed receiver can collect the diversity in both the delay and Doppler domains while simultaneously suppressing the negative impacts of DD-domain intersymbol interference (ISI) introduced by the doubly-selective fading channels.**

***Index Terms*—CP-free ODDM, local bi-orthgonal prototype pulse, cross-ambiguity function, delay-Doppler domain communications**

I. Introduction

Delay-Doppler (DD) domain communications, such as orthogonal time frequency space (OTFS) [2] and orthogonal delay-Doppler division multiplexing (ODDM) [3], are promising modulation schemes that can provide energy and spectral efficient wireless communications in high mobility environments. High speed movements of communication terminals lead to fast time variations of the wireless propagation environment and large Doppler spreads in wireless signals [4], [5]. It is extremely difficult to accurately estimate and track the fast time-varying channel directly in the time-frequency (TF) domain [6], [7]. In addition, time-selective fading due to higher Doppler spread introduces severe inter-carrier interference (ICI) in TF-domain multicarrier communication systems such as orthogonal frequency division multiplexing (OFDM) [5]. OTFS and ODDM systems address these challenges by modulating the information symbols directly in the DD domain, where the impulse response of the time and frequency doubly-selective channel changes much slower. Performing communications in the DD domain allows the systems to bypass challenging TF-domain operations while still enjoy the Doppler diversity in high mobility systems. Consequently, DD-domain communication systems are inherently superior to their TF-domain counterparts in high mobility systems owing to their robustness against large Doppler spreads [8].

The concept of DD-domain communications was first introduced in the landmark paper on OTFS [2], where the information symbols are modulated onto a two-dimensional DD-domain grid. Waveform design plays a critical role for OTFS systems. The ideal time-frequency bi-orthogonal waveforms, albeit unrealizable, have been believed as the lower bound on the OTFS error performance [2], [9]. The performance of OTFS systems with rectangular waveforms is evaluated in [9]. It is recently shown in [10] that the employment of practical bandlimited filter at the receiver can severely degrade the performance of OTFS with rectangular waveforms. Other widely-used pulse-shaping waveforms, such as prolate spheroidal waveforms, are adopted to reduce the out-of-band (OOB) emission and enable low complexity detections [11], [12]. To reduce complexity, many existing OTFS schemes follow an OFDM-based implementation [13], where the inverse discrete Fourier transform (IDFT) is used to approximate the continuous-time Wigner transform employed in the original OTFS architecture. The discrete nature of IDFT dictates that these works have to adopt a channel model with delays being integer multiples of the delay domain resolution [14], [15], which is not the case in a practical propagation environment.

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In addition, the IDFT-based implementation also requires extra modeling and processing to deal with the part of the Doppler spread that is a fractional multiple of the Doppler domain resolution [9], [16], and this incurs extra complexity in system designs and operations.

Even though OTFS performs modulation and detection in the DD domain, the prototype pulse and the corresponding matched filter are still designed and applied in the TF domain [2], [11], [17]. It is shown in [18] that the ideal TF-domain bi-orthogonal waveform does not necessarily result in a lower bound on OTFS error performance. This is due to the fact that the bi-orthogonality of the ideal waveforms in the TF domain is destroyed in the DD domain, and this causes severe DD-domain intersymbol interference (ISI). Despite the undesirable behaviors of the TF-domain waveforms in OTFS, they are chosen over their DD-domain counterparts mainly due to the Weyl-Heisenberg (WH) frame theory, which states that bi-orthogonal waveforms only exist if the product of the time (delay) and frequency (Doppler) resolution of the two dimensional TF-(DD-) domain modulation grid is no less than one [19], [20]. This WH condition can be generally met in the TF domain given the typical values of symbol period and subcarrier spacing in practical systems [21], [22]. However, given the fact that the delay is usually on the order of microsecond and the Doppler spread can be up to a few kilo-Hertz (KHz), the product resolution of the DD-domain modulation grid is usually several orders of magnitude less than one [23]. Thus based on the WH frame theory, there is no practical bi-orthogonal waveform in the DD domain.

ODDM systems bypass the restrictions of the WH frame theory by using a new concept of local bi-orthogonality [3], that is, a set of waveforms that are bi-orthogonal only within a small region on the DD-plane yet non-bi-orthogonal elsewhere. This does not violate the WH frame theory, which only dictates the existence of globally bi-orthogonal waveforms. In [23], [24], the ODDM waveforms are constructed through the repetition of a sequence of finite-length Nyquist elementary pulses, and they are directly applied to the information symbols in the DD domain. It is shown through theoretical analysis that the ODDM waveforms are locally bi-orthogonal inside an area with boundaries determined by the time and bandwidth constraints of a practical communication system [3], [23]. Through modulation with the locally bi-orthogonal waveform directly in the DD domain, the information symbols embedded in the ODDM waveform are mutually orthogonal with zero interference at the transmitter. The time-frequency dispersions of the doubly-selective channel still introduce interference at the receiver. However, the local bi-orthogonality leads to a highly structured system input-output (I/O) relationship, which can be used to significantly simplify the receiver complexity [25]–[27].

In [25], the performance of ODDM systems are studied with various receivers, including the message passing algorithm (MPA), iterative maximum-ratio combining (MRC) detector, and successive interference cancellation with minimum mean squared error (SIC-MMSE) detector. The I/O relationship of an IDFT-based ODDM system under a general physical channel is developed in [28], [29]. The employment of IDFT can reduce the implementation complexity. However, the adoption of IDFT implementation leads to an approximation of the original ODDM waveform.

This paper proposes a new CP-free ODDM system with practical prototype pulses under general doubly-selective fading channels. In the new ODDM system architecture, the communication receiver utilizes a receiving filter designed by temporally extending the ODDM prototype pulse, which is constructed through repetitions of finite length elementary pulses. The elementary pulse repetition at the transmitter and the temporal filter extension at the receiver introduce a wrap-around effect in the equivalent DD-domain channel spreading function without the need of using CP in the transmitted signal. The equivalent DD-domain channel spreading function incorporates the effects of various practical factors in the system, including the ODDM prototype pulse, the bandwidth and duration restrictions imposed by the DD-domain modulation grid, and the impulse response of the physical doubly-selective fading channel with arbitrary delays and Doppler spreads. We study the properties of the DD-domain spreading function by deriving the exact analytical expression of the cross-ambiguity function of the ODDM transmission and receiving waveforms. The main contributions of this paper lie in the following three aspects.

First, through theoretical studies of the cross-ambiguity function of the prototype pulses, it is analytically demonstrated that the elementary pulse repetition at the transmitter and temporal filter extension at the receiver cause a wrap-around effects in the equivalent DD-domain channel spreading function. The wrap-around effect leads to a block-circulant-like structure of the DD-domain channel matrix that is the same as conventional ODDM systems with CP. Thus the CP-free ODDM system can utilize the same receiver as conventional ODDM, yet it can achieve a higher energy efficiency because no energy needs to be allocated for CP.

Second, the ODDM system model describes the DD-domain I/O relationship by rigorously modeling a large number of factors and restrictions imposed by practical systems. Specifically, by considering the bandwidth and duration restrictions imposed by the DD-domain modulation grid, it can be used to model the behaviors of doubly-selective channels with arbitrary values of delays and Doppler spreads, and there is no need to normalize delays or Doppler spreads to integer multiples of the DD-domain grid size.

Third, a low complexity receiver is developed by exploiting the unique structure of the equivalent DD-domain channel matrix, which has a block-circulant-like structure thanks to the wrap-around effects of the equivalent DD-domain channel spreading function. The receiver performs interference cancellation and coherent matrix combining (CMC) in the delay domain, and minimum mean squared error (MMSE) detection in the Doppler domain. It can effectively suppress the DD-domain ISI while collecting the diversity in both the delay and Doppler domains.

The rest of the paper is organized as follows. Section II describes the system model of the new CP-free ODDM under practical doubly-selective fading channels with arbitrary delays and Doppler spreads. The system architecture and precise I/O modeling of CP-free ODDM are developed in Section III through theoretical analysis of the cross-ambiguity functions of the ODDM prototype pulses. In Section IV, a low complexity receiver is designed by exploiting the unique structure of the CP-free ODDM systems. Simulaton results are presented in Section V, and Section VI concludes the paper.

II. CP-Free ODDM System Model

The model of the CP-free ODDM systems is presented in this section. The DD domain is discretized into an $N \times M$ grid with N levels in the Doppler domain and M levels in the delay domain. The time duration of one ODDM frame is NT. The resolutions in the delay and Doppler domains are $T_s = \frac{T}{M}$ and $F_0 = \frac{1}{NT}$, respectively.

Denote $x[m,n]$ as the information symbol to be transmitted at delay mT_s and Doppler shift nF_0, for $m = 0, \cdots, M-1$ and $n = -\frac{N}{2}, \cdots, \frac{N}{2}-1$. The DD-domain information symbols are directly converted to a continuous-time time domain waveform as

$$x(t) = \sum_{m=0}^{M-1} \sum_{n=-N/2}^{N/2-1} x[m,n] g\left(t - mT_s\right) e^{j2\pi nF_0(t-mT_s)}, \quad (1)$$

where $g(t)$ is the ODDM prototype pulse defined as

$$g(t) = \sum_{\dot{n}=0}^{N-1} a(t - \dot{n}T), \quad (2)$$

with $a(t)$ being a square-root Nyquist elementary pulse parameterized by zero-ISI interval $T_s = \frac{T}{M}$. The energy of $a(t)$ is normalized to $\frac{1}{N}$, and the duration of $a(t)$ is limited to $T_a = QT_s \ll T$, with Q being an even integer.

Comment 1: The time domain ODDM signal in (1) is generated by directly applying the prototype pulse $g(t)$ to the DD domain signal $x[m,n]$. This is different from OTFS, where the DD-domain signal is first converted to the TF domain by using inverse symplectic finite Fourier transform (ISFFT), and the OTFS pulse shaping waveform is applied in the TF domain instead of the DD domain. Directly applying the waveform in the DD domain enables better controls the interference and diversity of the signals in the DD domain.

Comment 2: The DD-domain signal is placed on a two-dimensional (2D) lattice with coordinates (mT_s, nF_0). The lattice density is $\lambda = \frac{1}{T_s F_0} = MN \gg 1$. Based on the WH frame theory, when $\lambda \gg 1$, there exists no (bi)orthogonal waveform in the TF or DD domains. However, the ODDM waveform $g(t)$ defined in (2) satisfies local bi-orthogonality [23], that is, signals transmitted at different DD coordinates are mutually orthogonal on the discrete lattice grid points (mT_s, nF_0) for $|m| < M$ and $|n| < N$. Therefore, without the impact of the channel, there is no mutual interference among the symbols transmitted at different grid points on the DD domain lattice. Details will be elaborated in the next section.

The continuous-time signal is transmitted through a doubly-selective channel. It is assumed that the channel impulse response (CIR) is decided by a small number of reflectors. Hence, only a few parameters are needed to model the channel in the DD domain as

$$\varphi(\tau, \nu) = \sum_{i=1}^{P} \varphi_i \delta(\tau - \tau_i)\delta(\nu - \nu_i), \quad (3)$$

where $\delta(\cdot)$ is the Dirac delta function, φ_i, τ_i, and ν_i are the complex-valued path gain, delay, and Doppler shift for the i-th path, respectively, and P is the number of resolvable propagation paths. Without loss of generality, it is assumed that τ_i are in an ascending order and $\tau_1 = 0$. For channels with $\tau_1 > 0$ we can always introduce a delay in the received signal to make $\tau_1 = 0$. The signal observed at the receiver is

$$r(t) = \sum_{i=1}^{P} \varphi_i x(t - \tau_i) e^{j2\pi\nu_i(t-\tau_i)} + z(t), \quad (4)$$

where $z(t)$ is additive white Gaussian noise (AWGN) with one-sided power spectral density N_0.

The receiver employs a receiving filter that is designed as the temporally extended version of $g(t)$ as

$$p(t) = \sum_{\dot{n}=-1}^{N} a(t - \dot{n}T). \quad (5)$$

In $p(t)$, the elementary pulse $a(t)$ is repeated $N+2$ times by adding two additional elementary pulses to the prototype pulse $g(t)$, one on each end of $g(t)$. The temporal extension employed in $p(t)$ will cause a wrap-around effect in the equivalent DD domain channel spreading function. The wrap-around effect eliminates the need of CP in the transmitted signal, and details will be discussed in the next section through the theoretical analysis of the cross-ambiguity function between $p(t)$ and $g(t)$.

The received signal $r(t)$ passes through a bank of N receiving filters, $p^*(-t)e^{j2\pi f_k t}$, where $f_k = kF_0$ with $k = -\frac{N}{2}, \cdots, \frac{N}{2} - 1$. The output of the k-th receiving filter, $y(t, f_k) = r(t) \otimes p^*(-t)e^{j2\pi f_k t}$, with $\otimes$ being the convolution operator, can be written as

$$y(t, f_k) = \sum_{m=0}^{M-1} \sum_{n=-N/2}^{N/2-1} x[m,n] h_{m,n}(t, f_k) + w(t, f_k), \quad (6)$$

where

$$w(t, f) = z(t) \otimes p^*(-t)e^{j2\pi ft} \quad (7)$$

is the DD-domain noise components, and $h_{m,n}(t, f)$ is the equivalent DD-domain channel spreading function defined as

$$h_{m,n}(t, f) = e^{-j2\pi\frac{nm}{NM}} \sum_{i=1}^{P} \varphi_i e^{j2\pi(\nu_i + nF_0)(t-\tau_i)} \times A_{pg}(t - mT_s - \tau_i, f - nF_0 - \nu_i), \quad (8)$$

with

$$A_{pg}(t, f) = \int_{-\infty}^{\infty} p^*(\tau - t) g(\tau) e^{-2\pi f(\tau - t)} d\tau \quad (9)$$

being the cross-ambiguity function between $p(t)$ and $g(t)$. Detailed derivations of (6) can be found in Appendix A.

The output of the k-th matched filter is sampled at $t = lT_s + \tau_e$, where τ_e is used to model synchronization error between the transmitter and the receiver. It is assumed that $|\tau_e| \leq T_s$. This is a reasonable assumption as practical synchronization techniques can easily achieve sample level synchronizations. The discrete-time samples at the output of the k-th matched filter can then be represented as $y[l,k] = y(lT_s + \tau_e, kF_0)$,

$$y[l,k] = \sum_{m=0}^{M-1} \sum_{n=-N/2}^{N/2-1} x[m,n] h_{m,n}[l,k] + w[l,k], \tag{10}$$

where $w[l,k] = w(lT_s + \tau_e, kF_0)$, $h_{m,n}[l,k] = h_{m,n}(lT_s + \tau_e, kF_0)$ is the equivalent discrete DD-domain spreading function, and it can be calculated as

$$h_{m,n}[l,k] = e^{-j2\pi \frac{nm}{NM}} \sum_{i=1}^{P} \varphi_i e^{j2\pi(\nu_i + nF_0)(lT_s - (\tau_i - \tau_e))} A_{pg}((l-m)T_s - (\tau_i - \tau_e), (k-n)F_0 - \nu_i). \tag{11}$$

The discrete DD-domain channel spreading function $h_{m,n}[l,k]$ defined in (11) incorporates the impacts of the physical doubly-selective channel $\varphi(\tau, \nu)$, the prototype pulse $g(t)$, the synchronization error τ_e, and the temporally extended matched filter $p(t)$. The effects of the delay τ_i and Doppler shift ν_i of the physical channel are modeled by shifting the cross-ambiguity function, $A_{pg}(t,f)$, in the delay and Doppler domains, respectively.

As shown in (10), each information symbol, $x[m,n]$, is spread by $h_{m,n}[l,k]$ in both the delay domain l and Doppler domain k. That is, each transmitted information symbol is embedded in multiple received samples $\{y[l,k]\}_{l,k}$ due to the DD-domain spreading function $h_{m,n}[k,l]$, which renders both diversity and interference in the delay and Doppler domains. Larger delay spreads (or Doppler shifts) mean each information symbol is spread over a larger range in the delay (or Doppler) domain, and this leads to higher diversity orders, which can be quantified by analyzing the eigen-structure of the covariance matrix of the channel spreading functions [6].

Comment 3: With the equivalent discrete DD-domain system model defined in (10) and (11), the delay and Doppler shifts of the physical channel can take arbitrary values. It does not require the assumptions that the delay and/or the Doppler shift normalized with respect to their respective resolutions being an integer, or the need to distinguish between integer and fractional values of the normalized Doppler shift.

III. CP-Free ODDM System Architecture

The system architecture of the CP-free ODDM system is developed in this section through the analysis of the equivalent DD-domain channel spreading function $h_{m,n}[l,k]$ and the cross-ambiguity function $A_{pg}(t,f)$. The properties of $h_{mn}[l,k]$ depend on the behaviors of the cross-ambiguity function $A_{pg}(t,f)$, which plays an important role on the behavior and performance of the CP-free ODDM system.

A. Properties of the Cross-Ambiguity Function

The behavior of the DD-domain channel depends on the cross-ambiguity function $A_{pg}(t,f)$, which in turn depends on the ambiguity function of the elementary pulse $a(t)$.

Proposition III.1. *Assume the support of $a(t)$ is $[0, T_a]$ with $T_a = QT_s \ll \frac{T}{2}$. Then the cross-ambiguity function between $p(t)$ and $g(t)$ for $|t| < T + T_a$ is*

$$A_{pg}(t,f) = \sum_{k=0}^{N-1} e^{-j2\pi fkT} \times \begin{cases} A_a(t,f), & |t| \leq T_a, \\ A_a(t-T,f)e^{j2\pi Tf}, & |t-T| \leq T_a, \\ A_a(t+T,f)e^{-j2\pi Tf}, & |t+T| \leq T_a, \\ 0, & T_a < |t| < T - T_a, \end{cases} \tag{12}$$

where

$$A_a(t,f) = \int_{-\infty}^{\infty} a^*(\tau - t) a(\tau) e^{-j2\pi f(\tau - t)} d\tau \tag{13}$$

is the ambiguity function of the elementary pulse $a(t)$.

Proof. The proof is in Appendix B. ■

Proposition III.1 describes the relationship between the cross-ambiguity function $A_{pg}(t,f)$ and the elementary pulse ambiguity function $A_a(t,f)$. Given the fact that the prototype pulse $g(t)$ is constructed by repeating the elementary pulse $a(t)$ N times in the time domain with spacing T as in (2), the cross-ambiguity function $A_{pg}(t,f)$ can be obtained by repeating the elementary ambiguity function $A_a(t,f)$, up to a phase shift, within $|t| < T + T_a$ in the time domain with spacing T. Since the equivalent DD-domain spreading function $h_{mn}[l,k]$ is a superposition of shifted versions of $A_{pg}(t,f)$, the repetition in the cross-ambiguity function will be manifested in the equivalent DD-domain spreading function. The repetition behavior in $A_{pg}(t,f)$ will cause a wrap-around effect of the equivalent DD-domain spreading function in the delay domain. We have the following corollary regarding the delay domain behavior of $h_{m,n}[l,k]$.

Corollary III.1. *Assume $2(Q+1) + \lfloor \frac{\tau_{\max}}{T_s} \rfloor < M$ with $\tau_{\max} = \max_i(\tau_i)$ being the maximum delay spread. The DD-domain spreading function defined in (11) has a finite support in the delay domain. That is, $h_{m,n}[l,k] \neq 0$ for $l - m \in \mathcal{L}$, where*

$$\mathcal{L} = -M + [1, L_2] \cup [-L_1, L_2] \cup M - [L_1, 1], \tag{14}$$

with

$$L_1 = Q + 1 \text{ and } L_2 = Q + 1 + \lfloor \frac{\tau_{\max}}{T_s} \rfloor. \tag{15}$$

Proof. The proof is in Appendix C. ■

The assumption $2(Q+1) + \lfloor \frac{\tau_{\max}}{T_s} \rfloor < M$ is used in the above corollary. This is a mild assumption given that the value of T is usually chosen such that $\tau_{\max} \ll T$ and $Q \ll M$. The delay domain support is non-causal due to the non-causal definition of the matched filter $p^*(-t)e^{j2\pi ft}$. The matched filter and the spreading function can always be made causal by introducing a delay in the matched filter.

Comment 4: The delay domain support of $h_{mn}[l,k]$ is divided into three sections. The two sections on the edges, $-[M-1, M-L_2]$ and $[M-L_1, M-1]$, correspond to the supports of $A_{pg}(t,f)$ for $|t+T| \leq T_a$ and $|t-T| \leq T_a$, respectively. These two edge sections introduce a wrap-around in the delay domain, which will create an effect that is similar to cyclic prefix and cyclic postfix. As a result, there is no need for cyclic prefix or cyclic postfix for the proposed ODDM architecture. All-zero guard intervals between ODDM frames are still needed to avoid interference between adjacent frames.

Next we study the bi-orthogonality behavior of the pulses $p(t)$ and $g(t)$ in the DD domain.

Corollary III.2. *The pulses $p(t)$ and $g(t)$ satisfy local bi-orthogonality for $|t| < T$ and $|f| < \frac{1}{T}$, that is*

$$A_{pg}(mT_s, nF_0) = \delta[m]\delta[n], \text{ for } |m| < M, |n| < N. \quad (16)$$

Proof. The proof is in Appendix D. ∎

Fig. 1 shows the cross-ambiguity function $A_{pg}(t,f)$ with the elementary pulse $a(t)$ being a rectangular pulse with pulse width T_s. The parameters are set at $N = 8$ and $M = 4$, which correspond to a delay domain support $|t| < 4T_s = T$ and Doppler domain support $|f| < 8F_0 = \frac{1}{T}$. The DD-domain grid points with coordinates $\{(mT_s, nF_0)\}_{|m|<M,|n|<N}$ are shown as red dots in the figures. The results illustrate that the cross-ambiguity function satisfies local bi-orthogonality for $|t| < T$ and $|f| < \frac{1}{T}$, that is, its value is 0 at grid coordinates (mT_s, nF_0) when $mn \neq 0$, $|m| < M$ and $|n| < N$.

Comment 5: Based on the WH frame theory, no practical bi-orthogonal waveform exists over a DD-domain grid with the delay-Doppler resolution (T_s, F_0), because the product of the delay and Doppler resolutions is $T_sF_0 = \frac{1}{MN} \ll 1$. However, Corollary III.2 demonstrates that the prototype pulse $g(t)$ and $p(t)$ can achieve *local bi-orthogonality* over the region $|t| < T$ and $|f| < \frac{1}{T}$, which coincides with the region over which the information symbols are transmitted. Even though the waveforms are not bi-orthogonal outside this region, the non-orthogonality has no impact on the transmitted symbols in the ODDM system.

Comment 6: Since the pulses $p(t)$ and $g(t)$ are locally bi-orthogonal over the grid with delay-Doppler resolution (T_s, F_0), the information symbols embedded in the continuous-time signal $x(t)$ at the transmitter are mutually orthogonal. That is, there is no ISI in the transmitted signal. However, due to the effects of doubly-selective fading, each information symbol is dispersed in both the delay and Doppler domains as in the DD-domain system equation (10). Based on the definition of the DD-domain spreading function $h_{mn}[l,k]$ defined in (11), the information symbol $x[m,n]$ is dispersed in both the delay and Doppler domains by $h_{m,n}[l,k]$. The dispersions introduce both diversity and interference in the delay and Doppler domains.

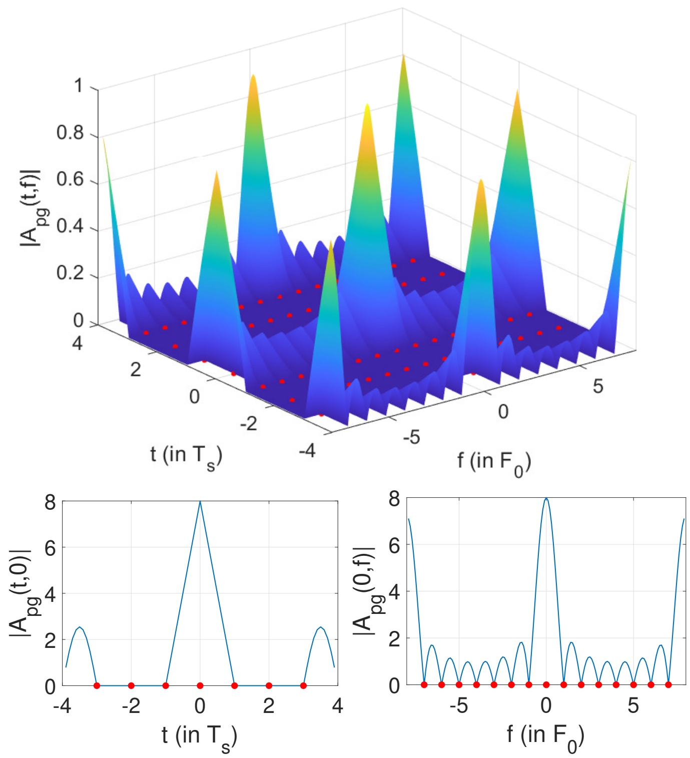


Fig. 1: Cross-ambiguity function of $p(t)$ and $g(t)$ with rectangular elementary pulses ($N = 8$, $M = 4$). The grid points are marked with red dots.

B. Matrix Representation in the DD Domain

The DD-domain system equation in (10) can be written in a matrix format as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}, \quad (17)$$

where

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_M \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_M \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_M \end{bmatrix}, \quad (18)$$

are length NM vectors, with $\mathbf{y}_m = [y[m, -N/2], \cdots, y[m, N/2-1]]^T \in \mathcal{C}^{N\times 1}$, $\mathbf{x}_m = [x[m, -N/2], \cdots, x[m, N/2-1]]^T \in \mathcal{C}^{N\times 1}$, and $\mathbf{w}_m = [w[m, -N/2], \cdots, w[m, N/2-1]]^T \in \mathcal{C}^{N\times 1}$ being length N vectors. The equivalent DD-domain channel matrix $\mathbf{H} \in \mathcal{C}^{NM\times NM}$ can be expressed as

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{00} & \mathbf{H}_{01} & \cdots & \mathbf{H}_{0(M-1)} \\ \mathbf{H}_{10} & \mathbf{H}_{11} & \cdots & \mathbf{H}_{1(M-1)} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{H}_{(M-1)0} & \mathbf{H}_{(M-1)1} & \cdots & \mathbf{H}_{(M-1)(M-1)} \end{bmatrix}, \quad (19)$$

where $\mathbf{H}_{lm}$ is an $N \times N$ matrix with the $(k+1, n+1)$-th element being $h_{m,n-\frac{N}{2}}[l, k-\frac{N}{2}]$, for $l, m = 0, \cdots, M-1$, and $k, n = 0, \cdots, N-1$.

Since the equivalent DD-domain spreading function defined in (11) has a finite support as shown in Corollary III.1, the DD-domain channel matrix $\mathbf{H}$ will be sparse with many blocks in (19) being all-zero matrices. Specifically, the DD-domain channel matrix $\mathbf{H}$ has a block-circulant-like structure as stated in the following proposition.

Proposition III.2. *The DD-domain channel matrix $\mathbf{H}$ has a block-circulant-like structure, with $\mathbf{H}_{lm}$ being an all-zero matrix if $l - m \notin \mathcal{L}$, where $\mathcal{L}$ is defined in* (14)*, and $l, m = 0, \cdots, M-1$.*

Proof. The proof is in Appendix E. ■

An example of $\mathbf{H}$ with $M = 16$, $L_1 = 3$ and $L_2 = 5$ are shown in Fig. 2. Each sub-matrix $\mathbf{H}_{lm}$ corresponds to the delay $(l-m)T_s$, and its element $h_{m,n}[l,k]$ corresponds to the Doppler shift at $(k-n)F_0$. Based on Proposition III.2 and Corollary III.1, on the l-th row block of $\mathbf{H}$, we have $\mathbf{H}_{lm} \neq 0$, if $l - m \in \mathcal{L} = [-15, -11] \cup [-3, 5] \cup [13, 15]$. Each row block of $\mathbf{H}$ has $L = L_1 + L_2 + 1 = 9$ non-zero sub-matrices, which correspond to a channel length of $L = 9$ in the delay domain.

The circulant-like wrap-around of the non-zero sub-matrices in the upper-right and lower-left corners of $\mathbf{H}$ are caused by the temporal repetitions of the elementary waveform $a(t)$ that is unique to the ODDM prototype pulse $g(t)$. Specifically, the upper-right corner corresponds to the support of $A_{pg}(t, f)$ with $t \in -T + [-T_a, T_a]$, which is mapped to a delay domain support of $l - m \in -M + [1, L_2] = [-15, -11]$ for $\mathbf{H}_{lm}$. Similarly, the lower-left corner corresponds to the support of $A_{pg}(t, f)$ with $t \in T + [-T_a, T_a]$, which is mapped to a delay domain support of $l - m \in M - [L_1, 1] = [13, 15]$.

Comment 7: The temporal extension of the receiving filter introduces a wrap-around in the delay domain, which creates an effect that is equivalent to the cyclic prefix. As a result, even though there is no cyclic prefix added at the transmitter, the structure of the signal at the output of the receiving filter is equivalent to that of a conventional ODDM system with CP. That is, the circulant-like wrap-around structure of $\mathbf{H}$ shown in Fig. 2 is similar to that of an ODDM system with CP [3]. Therefore, similar receivers can be used for both CP-free ODDM and CP-ODDM systems without performance penalty.

Even though no CP is needed, we still need to add all-zero guard intervals (GI) before and after each ODDM frame to avoid inter-frame interference. Based on the delay-domain support of $h_{m,n}(l,k)$ in Corollary III.1, the length of the GI before and after each ODDM frame should be $L_2 N$ and $L_1 N$, respectively. Since all-zero sequences instead of CP are used for the GI, all energy can be used for the actual data symbols and there is no need to allocate energy for CP transmissions. This leads to a higher energy efficiency as compared to conventional ODDM and OTFS systems.

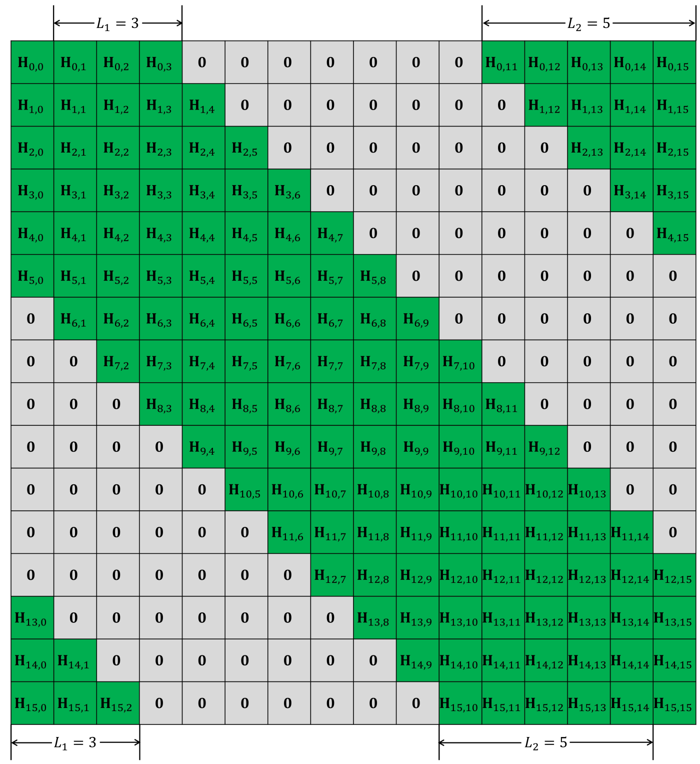


Fig. 2: Delay-Doppler domain channel matrix with $M = 16$, $L_1 = 3$, and $L_2 = 5$.

C. CP-Free ODDM Waveform Properties

The out-of-band emission (OOBE) and peak-to-average power ratio (PAPR) of the CP-free ODDM waveforms are studied in this subsection.

OOBE and PAPR are two important metrics for evaluating practical communication waveforms. A lower OOBE means less energy leakage out of the assigned channel, thus less interferences to other users and a higher spectral efficiency. The OOBE can be measured by using adjacent channel power ratio (ACPR), which is defined as the ratio between the power of the adjacent channels and that of the main spectral lobe. A lower ACPR means a lower OOBE. PAPR is defined as the ratio between the peak power to the average power in the time domain waveforms. A lower PAPR means less power fluctuations in the time domain waveform, and it allows the employment of non-linear power amplifiers, which in general has a higher energy efficiency than their linear counterparts.

Table 1 compares the ACPR and PAPR of the proposed CP-free ODDM system with those of the conventional ODDM systems with CP (labeled as CP-ODDM) [3], under various practical elementary waveforms, including the rectangular, sinc, and root raised cosine (RRC) waveforms. The sinc and RRC waveforms are truncated with parameter $Q = 8$. The results demonstrate that the CP-free ODDM systems share similar ACPR and PAPR as CP-ODDM systems. The PAPR of the CP-free ODDM system is slightly lower than those of CP-ODDM system for all three prototype pulses; the ACPR of the CP-free

Table 1: ACPR and PAPR (in dB)

Pulse	CP-ODDM		CP-free ODDM	
	ACPR	PAPR	ACPR	PAPR
Rectangular	-11.83	13.92	-12.07	13.07
Sinc	-18.50	14.77	-18.17	13.71
RRC ($\alpha = 1$)	-46.96	13.92	-46.65	13.00

ODDM is slightly lower than that of the CP-ODDM with the rectangular elementary pulse, and the trend is reversed for sinc and RRC elementary pulses. The differences among all cases are marginal. Thus employing all-zero sequences instead of CP will not have significant impacts on the OOBE and PAPR of the CP-free ODDM waveforms. Among the three elementary pulses, RRC pulse with a roll-off factor $\alpha = 1$ has the best performance in terms of both ACPR and PAPR.

IV. Low Complexity Iterative Receiver

Based on the block-circulant-like structure of the DD-domain channel matrix as shown in Proposition III.2 and Fig. 2, we design a low complexity iterative receiver that can recover the information symbols in the DD domain.

A. Receiver Design

In the proposed receiver, we perform coherent matrix combining (CMC) and successive interference cancellation (SIC) in the delay domain, and minimum mean squared error (MMSE) detection in the Doppler domain.

From (17) and Proposition III.2, the received samples corresponding to the n-th DD-domain block, $\mathbf{x}_n$, can be written as

$$\mathbf{y}_l = \mathbf{H}_{ln}\mathbf{x}_n + \sum_{\substack{m\neq n\\ l-m\in\mathcal{L}}} \mathbf{H}_{lm}\mathbf{x}_m + \mathbf{w}_l, \text{ for all } l-n\in\mathcal{L}. \quad (20)$$

Due to the block-circulant-like structure of $\mathbf{H}$, the n-th DD domain data block, $\mathbf{x}_n$, will be embedded in a subset of the DD domain received samples, $\{\mathbf{y}_l\}_{l-n\in\mathcal{L}}$. In the mean time, $\mathbf{x}_n$ will be interfered by a subset of data blocks, $\mathbf{x}_m$, for all $l-m\in\mathcal{L}$ and $m\neq n$.

The system model in (20) is used to detect $\mathbf{x}_n$, and $\{\mathbf{x}_m\}_{l-m\in\mathcal{L},m\neq n}$ are treated as interference. During the iterative detection algorithm, each iteration contains three steps as follows.

- **Step 1: Interference Cancellation.** During the detection of $\mathbf{x}_n$, interference cancellation is performed in the delay domain as

$$\begin{aligned}\hat{\mathbf{y}}_l &= \mathbf{y}_l - \sum_{\substack{m\neq n\\ l-m\in\mathcal{L}}} \mathbf{H}_{lm}\hat{\mathbf{x}}_m \\ &\approx \mathbf{H}_{ln}\mathbf{x}_n + \mathbf{w}_l, \text{ for all } l-n\in\mathcal{L},\end{aligned} \quad (21)$$

  where $\hat{\mathbf{x}}_m$ is the previously detected symbol vector, and it includes symbols detected before $\mathbf{x}_n$ in both the same and previous iterations. The assumption of perfect interference cancellation is used in the second equality. It should be noted the assumption of perfect interference cancellation is only used to facilitate receiver design, and actual interference cancellation with potential detection errors are used in practical receiver implementations in all simulations. It will be shown in simulations that the assumption of perfect interference cancellation leads to reasonably good results especially at higher signal-to-noise ratio (SNR) after a few iterations of the proposed algorithm.
- **Step 2: Coherent Matrix Combining.** After interference cancellation, performing CMC over $\{\hat{\mathbf{y}}_l\}_{l-n\in\mathcal{L}}$ as

$$\boldsymbol{\gamma}_n = \sum_{l-n\in\mathcal{L}} (\mathbf{H}_{ln})^H \hat{\mathbf{y}}_l. \quad (22)$$

  After CMC, the DD-domain system equation in (20) can be written as

$$\boldsymbol{\gamma}_n = \boldsymbol{\Lambda}_n\mathbf{x}_n + \boldsymbol{\varepsilon}_n, \quad (23)$$

  where

$$\begin{aligned}\boldsymbol{\Lambda}_n &= \sum_{l-n\in\mathcal{L}} (\mathbf{H}_{ln})^H \mathbf{H}_{ln}, \\ \boldsymbol{\varepsilon}_n &= \sum_{l-n\in\mathcal{L}} (\mathbf{H}_{ln})^H \mathbf{w}_l.\end{aligned} \quad (24)$$

- **Step 3: Doppler Domain MMSE.** After SIC and CMC in the delay domain, the system equation in (23) contains the information regarding $\mathbf{x}_n$ in the Doppler domain. Specifically, the (u, v)-th element of the equivalent channel matrix $\boldsymbol{\Lambda}_n$ corresponds to the Doppler shifts at $(u-v)F_0$. We propose to detect $\mathbf{x}_n$ from $\boldsymbol{\gamma}_n$ by following the MMSE criterion as

$$\tilde{\mathbf{x}}_n = \mathbf{W}_n\boldsymbol{\gamma}_n, \quad (25)$$

  where $\mathbf{W}_n$ is designed to minimize the mean squared error (MSE) $\mathbb{E}[\|\tilde{\mathbf{x}}-\mathbf{x}\|^2]$. Based on the orthogonality principle,

$$\mathbb{E}[(\mathbf{W}_n\boldsymbol{\gamma}_n - \mathbf{x}_n)\boldsymbol{\gamma}_n^H] = \mathbf{0}, \quad (26)$$

  the MMSE matrix $\mathbf{W}_n$ can be solved as

$$\mathbf{W}_n = \boldsymbol{\Lambda}_n^H\left(\boldsymbol{\Lambda}_n\boldsymbol{\Lambda}_n^H + \frac{N_0}{E_s}\boldsymbol{\Lambda}_n\right)^{-1}. \quad (27)$$

In the proposed iterative algorithm, in each iteration, the data in the n-th DD domain data block, $\tilde{\mathbf{x}}_n$, is detected by first performing SIC as in (21), then CMC as in (22), and finally MMSE detection by using (25) and (27), for $n = 1, \cdots, N$. The detected symbols are hard detected to the modulation constellation as $\{\hat{\mathbf{x}}_n\}_{n=1}^N$, then used for SIC for the detection of the subsequent symbols, including symbols within the same iteration. The inputs to the first iteration are assumed to be 0. The iteration terminates when the output of two consecutive iterations are the same, or the maximum number of iterations is reached.

The CMC-MMSE receiver requires the knowledge of the channel state information (CSI). Given that the equivalent DD-domain channel transfer function changes much slower that its TF-domain counterpart, channel estimations in ODDM systems are usually fairly accurate [8], [10]. Similar to many existing

Table 2: Number of complex multiplications for proposed and traditional detection algorithms.

Algorithm	Complex Multiplication Count
MPA	$N_{\text{iter}} M N^3 (L_1 + L_2)^2 (2\lvert\mathcal{S}\rvert^2 + 4\lvert\mathcal{S}\rvert + 1)$
MMSE	$\frac{4}{3} M^3 N^3 + 2M^2 N^2 - MN$
CMC-MMSE (Proposed)	$N_{\text{iter}} M \big(N^3(\frac{4}{3} + L_1 + L_2 + 1)$ $+ N^2(2(L_1 + L_2) + 3) - N\big)$

works [3], [9]–[12], [16], it is assumed in this paper that the receiver has perfect knowledge of the CSI.

B. Complexity Analysis

The complexity of the proposed CMC-MMSE receiver is evaluated and compared to existing receivers in the literature. The complexity is measured by using the number of complex multiplications or the simulation run time during the detection of a single frame.

Table 2 shows the number of complex multiplications required for detecting the information in one frame by the MPA, MMSE, and CMC-MMSE receivers, respectively, where N_{iter} is the maximum number of iterations employed by the MPA or CMC-MMSE receiver. The MMSE receiver is directly deployed for the $NM \times NM$ system in (17), thus it requires the inversion of a size-$(NM \times NM)$ matrix. The CMC-MMSE algorithm decomposes the $NM \times NM$ system into multiple $N \times N$ sub-systems as in (20), where interference cancellation is performed among the sub-systems, and MMSE only needs to be performed for the sub-systems. Such a system decomposition leads to significant complexity reduction. As can be seen from the table, the complexity of the MMSE algorithm is dominated by $\mathcal{O}(\frac{4}{3} M^3 N^3)$ due to the size-$(NM \times NM)$ matrix inversion, while the dominating factor of the CMC-MMSE algorithm is $\mathcal{O}\left(M N_{\text{iter}} (\frac{4}{3} + L_1 + L_2 + 1) N^3\right)$. Simulation results demonstrate that the CMC-MMSE algorithm usually converges with $N_{\text{iter}} \leq 3$ iterations. it addition, for practical channels we have $L_1 + L_2 \ll M$. Consequently, the complexity of the MMSE algorithm grows in $(MN)^3$ yet that of CMC-MMSE scales with $N_{\text{iter}} M N^3 (L_1 + L_2)$. The complexity of the MPA algorithm grows with $N_{\text{iter}} M N^3 (L_1 + L_2)^2 \lvert\mathcal{S}\rvert^2$, where $\lvert\mathcal{S}\rvert$ is the cardinality of the modulation constellation $\mathcal{S}$. In addition, the number of iterations required by MPA is usually at least one order of magnitude larger than that for the CMC-MMSE algorithm. Thus the complexity of CMC-MMSE is much lower than that of the MMSE and MPA algorithms.

The above analysis is corroborated by the results shown in Fig. 3, which compares the average simulation run time per frame required by the various detection algorithms. The simulation is performed by using Matlab R2024b on a Windows 10 workstation equipped with Intel i7 7700K CPU and 32 GB of RAM. The simulation employs the RRC filter with a roll-off factor $\alpha = 0.4$ and a truncation parameter $Q = 8$. The average run time of the MPA algorithm is at least one magnitude higher than that of the MMSE or CMC-MMSE algorithms, mainly due to the large number of iterations needed for the MPA algorithm to converge. For a fixed N, the average run time of the MMSE

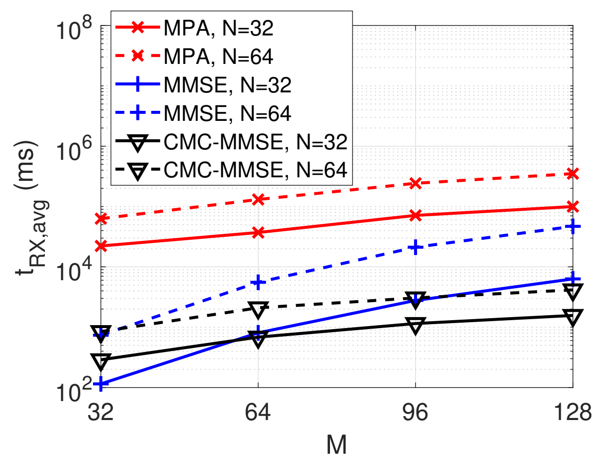


Fig. 3: Average simulated run time for typical receivers and proposed CMC-MMSE.

algorithm scales with high order exponent of M, yet that of the proposed CMC-MMSE algorithm scales linearly with M. The results demonstrate that the proposed CMC-MMSE algorithm has the lowest complexity among the three.

V. Simulation Results

The performance of the proposed CP-free ODDM system is evaluated in this section. The carrier frequency is set at 4 GHz. The value of T is set such that $\frac{1}{T} = 15$ KHz. The physical channel is generated with an EVA delay profile [30] under various speeds of the user equipment (UE). In all simulations, the modulation is QPSK. Unless specified otherwise, RRC filters with a roll-off factor $\alpha = 0.4$ and truncation length $Q = 8$ is used during the simulations.

We first study the convergence of the CMC-MMSE algorithm in Figure 4, where the bit error rate (BER) of the CP-free ODDM system is shown as a function of the number of iterations during detection. During the simulations, we set $M = 64$ and $E_b/N_0 = 18$ dB. Under all system configurations, large performance improvements are achieved during the first 3 iterations, and the iterative processes converge after 3 or 4 iterations. Based on these results, the maximum number of iterations of the CMC-MMSE receiver is set at $N_{\text{iter}} = 3$ for all subsequent simulations.

Figure 5 compares the BER performance of the proposed CP-free ODDM system with those of the CP-free OFDM [31], CP-OTFS [17], and CP-ODDM systems [3], at a UE speed of 500 km/hr. For fair comparison, the CP-free OFDM system operates with $N \times M = 1024$ subcarriers, which result in the same number of information symbols per frame as the DD-domain systems. he CMC-MMSE receiver is used for both CP-free ODDM and CP-ODDM systems. The performance of the CP-free OFDM system is considerably worse than that of the DD-domain systems. The performance degradation in OFDM is mainly due to the significant amount of ICI introduced by the fast time variation of the doubly-selective fading channel. The DD-domain systems, on the other hand, can effectively exploit

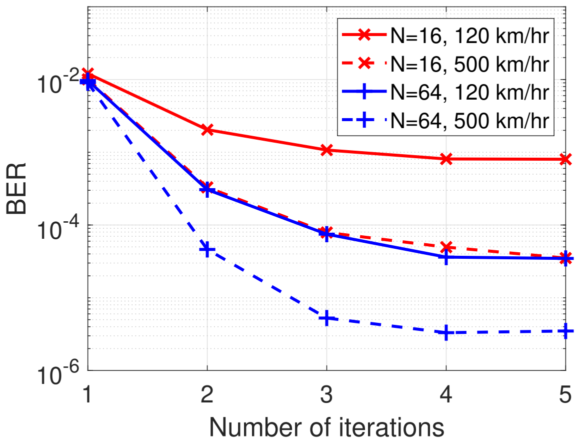


Fig. 4: BER performance of CMC-MMSE v.s the number of maximum iterations ($E_b/N_0 = 18$ dB).

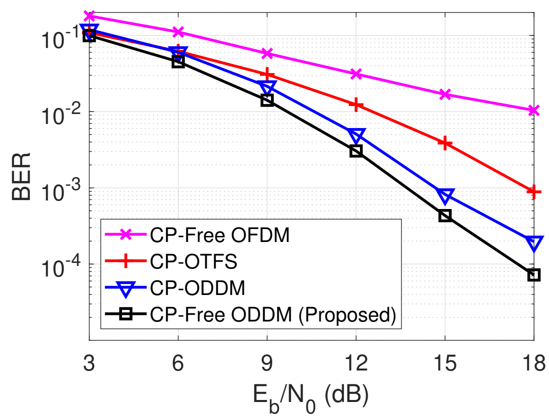


Fig. 5: BER performance of OFDM, OTFS and ODDM systems ($N = 16$, $M = 64$, 500 km/hr).

the Doppler diversity in high mobility scenarios. The proposed CP-free ODDM system consistently outperforms the CP-OTFS and CP-ODDM systems, and the performance improvement is mainly contributed by the energy savings from the removal of the CP. At BER = 10^{-3}, the CP-free ODDM outperforms the CP-ODDM and CP-OTFS systems by 1.0 dB and 4.1 dB, respectively.

Figure 6 compares the spectral efficiency of CP-free ODDM and CP-ODDM systems by using the metric of system throughput. The throughput is evaluated as $\log_2(|\mathcal{S}|)(1 - \text{FER})\frac{NM}{NM+N_{\text{CP}}}$, where FER is the frame error rate evaluated through simulations, $|\mathcal{S}|$ is the cardinality of the modulation constellation set $\mathcal{S}$, and N_{CP} is the number of CP symbols for CP-ODDM or the number of zero-padding symbols for CP-free ODDM. Both systems employ the CMC-MMSE receiver. The proposed CP-free ODDM system achieves a higher throughput than its CP counterpart. Thus, the CP-free ODDM system outperforms CP-ODDM in terms of both energy efficiency as in Figure 5 and spectral efficiency as in Figure 6.

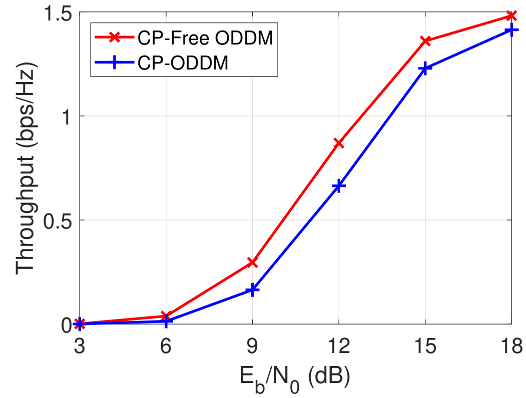


Fig. 6: Throughput of CP-ODDM and CP-Free ODDM ($N = 16$, $M = 64$, 500 km/hr).

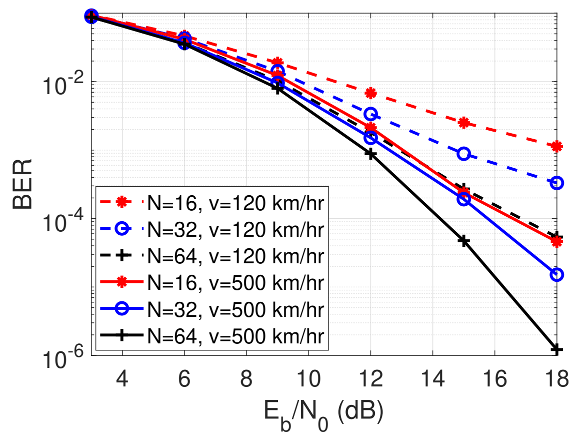


Fig. 7: BER performance of ODDM under various values of N and UE speed ($M = 64$).

Figure 7 shows the BER of CP-free ODDM systems under various values of N and UE speeds. The value of M is fixed at 64. The BER performance improves as N becomes larger. The performance gain is contributed by the higher Doppler diversity order with a longer ODDM frame duration NT. In addition, the CP-free ODDM system can effectively collect the Doppler diversity in high mobility systems. At BER = 10^{-4}, the high mobility system with $N = 64$ operating at a UE speed of 500 km/hr outperforms its 120 km/hr counterpart by 2.4 dB, and the gap grows larger at lower BER levels.

Figure 8 compares the performance of CP-free ODDM systems with different elementary pulse shaping waveforms $a(t)$. The rectangular waveform has a finite time support with $Q = 1$. The sinc and RRC waveforms are truncated in the time domain with $Q = 8$. The simulation parameters are set as $N = 64$, $M = 64$, and UE speed at 500 km/hr. Among the four cases considered in this example, the RRC pulse with a roll-off factor $\alpha = 0.4$ and $Q = 8$ achieve the best performance, followed by the RRC waveform with $\alpha = 1$ and the rectangular waveform. The sinc waveform initially outperforms the RRC

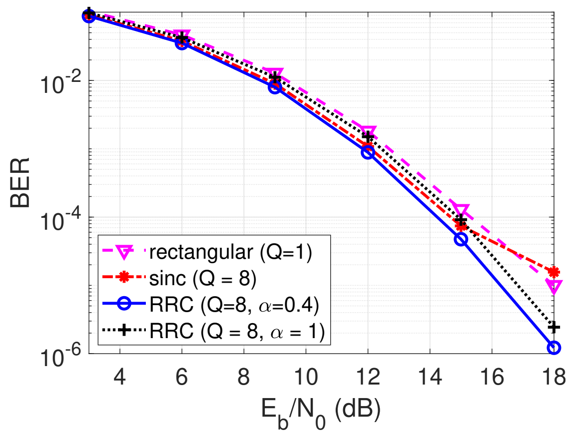


Fig. 8: BER performance of ODDM with different elementary pulses ($N = 64, M = 64$).

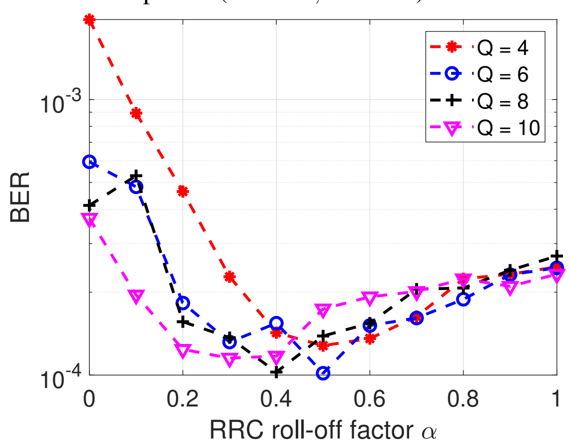


Fig. 9: Impacts of truncation length Q and roll-off factor α of the RRC elementary pulse ($N = 16, M = 64$).

waveform with $\alpha = 1$, but its performance drops significantly at $E_b/N_0 = 18$ dB. This is due to the fact that the finite truncation length of the sinc waveform causes non-negligible spectral energy leakage, and it also destroys its Nyquist property. Consequently, this introduces interference in both the delay and Doppler domains. At higher E_b/N_0 the system performance is dominated by the interference, which causes performance degradation for systems with the truncated sinc waveform. Among the four systems, the system with the rectangular waveform has the largest bandwidth. Thus the truncated RRC waveform can achieve a better spectral efficiency at a lower BER.

To further study the impacts of the truncation length Q and roll-off factor α of the elementary RRC waveforms, we plot the BER as a function of α in Figure 9 under different values of truncation length Q. The simulation parameters are set at $N = 16$, $M = 64$, $v = 500$ km/hr, and $E_b/N_0 = 16$ dB for all curves. For a fixed Q, the BER performance is in general a quasi-convex function of the roll-off factor α. The BER performance variation is due to the tradeoff between diversity and interference. When α is small, the RRC waveform has a relative larger tail in the delay domain, and a shorter truncation length with a smaller Q will cause larger energy leak thus strong interference in the Doppler domain. Consequently, the

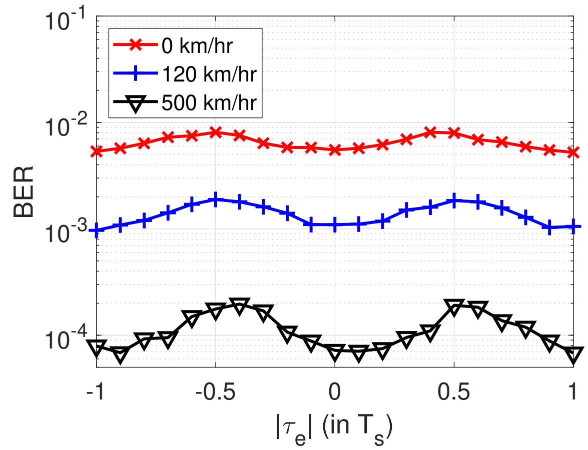


Fig. 10: Impacts of timing offset τ_e on ODDM performance ($N = 16,\ M = 64$).

BER performance improves as Q becomes larger at $\alpha = 0$ (sinc waveform). When α is large (e.g. $\alpha = 1$), the delay domain tail of the RRC waveform diminishes rapidly, thus the performance difference among the systems with different Q is very small. A larger Q will result in a longer channel length, which helps collect the diversity in the delay domain, but at the cost of a higher interference level. A larger α allows for shorter truncation length but will lead to higher interference in the Doppler domain. Under the current system configuration, the best overall performance is achieved by setting $(Q = 8, \alpha = 0.4)$ or $(Q = 6, \alpha = 0.5)$.

Figure 10 studies the impacts of synchronization error τ_e on the performance of the CP-free ODDM system. At lower speeds, synchronization timing offset causes small performance deterioration, but the effect is amplified at higher UE speeds. It is interesting to note that perfect synchronization with $\tau_e = 0$ does not necessarily mean the best performance. For example, at 500 km/hr, the best performance is achieved at $\tau_e = 0.1T_s$. This result agrees with those presented in [32], which analytically shows that the error performance of a single carrier system is a periodic function in τ_e, and the timing offset that achieves the best performance depends on the power delay profile of the channel.

Figure 11 compares the BER performance of the CP-free ODDM system with different low complexity receivers, including the MPA, the MMSE, and the proposed CMC-MMSE receivers. The UE speed is 120 km/hr. High error floors are observed for systems with the MPA receivers. This is mainly caused by the large number of loops in the bi-partite graph of the DD domain channel matrix. It is well known that the MPA receiver can operate effectively on bi-partite graphs without loop, but its performance deteriorates significantly with the presence of loops, which cause positive feedback during the iterative detection process. Both MMSE and CMC-MMSE receivers outperform the MPA receiver. When $N = 16$, the CMC-MMSE and MMSE receivers achieve a similar performance, but the CMC-MMSE receiver outperforms the MMSE receiver

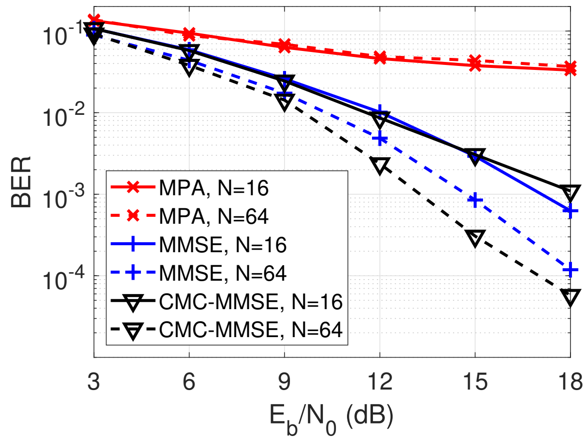


Fig. 11: BER performance of various receivers. (120 km/hr).

at $N = 64$. When N is large, the linear MMSE receiver is no longer able to handle the large amount of interferences in the delay and Doppler domains. In addition, as shown in Fig. 3, the CMC-MMSE has the lowest complexity among the three receivers. Thus the proposed CMC-MMSE receiver can achieve a better BER performance at a lower complexity.

VI. Conclusions

We have developed a new CP-free ODDM system operating in general doubly-selective fading channels. The CP-free architecture was made possible by exploiting the unique structures of the ODDM prototype pulses. It has been shown that the elementary pulse repetition at the transmitter along with the temporal filter extension at the receiver introduces a wrap-around effect in the equivalent channel spreading function. The wrap-around effect leads to a block-circulant-like structure of the DD-domain channel matrix that is the same as conventional ODDM systems with CP. The CP-free ODDM system model incorporates the effects of practical ODDM prototype pulses and general doubly-selective fading channel with arbitrary delays and Doppler shifts. It does not require many of the impractical assumptions adopted by existing OTFS studies, such as channel delay being integer multiple of the delay domain resolution, or the prototype pulse is localized in both the delay (time) and Doppler (frequency) domains. Furthermore, the unique structure of the equivalent DD-domain channel matrix has enabled the design of a low complexity CMC-MMSE iterative receiver. Simulation results have demonstrated that the CP-free ODDM system outperforms existing systems in terms of both energy efficiency and spectral efficiency.

Appendix A
Derivations of DD-Domain I/O Relationship in (6)

The output of the receiving filter in the DD domain is the convolution between $r(t)$ and the receiving filter $p^*(-t)e^{j2\pi f_k t}$ as

$$y(t, f_k) = \int r(t')p^*(t'-t)e^{-j2\pi f_k(t'-t)}dt'. \tag{28}$$

The received signal minus the noise term can then be calculated as

$$\begin{aligned}
&y(t, f_k) - w(t, f_k) = \\
&\sum_{i=1}^{P}\varphi_i \int p^*(t'-t)x(t'-\tau_i)e^{-j2\pi f_k(t'-t)}e^{j2\pi\nu_i(t'-\tau_i)}dt', \\
&= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{i=1}^{P}\varphi_i x[m,n]\int p^*(t'-t)g\left(t'-\tau_i-mT_s\right) \\
&e^{j2\pi nF_0\left(t'-\tau_i-mT_s\right)}e^{-j2\pi f_k(t'-t)}e^{j2\pi\nu_i(t'-\tau_i)}dt', \\
&= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{i=1}^{P}\varphi_i x[m,n]\int p^*(\tau-(t-\tau_i-mT_s))g\left(\tau\right) \\
&e^{j2\pi nF_0\tau}e^{-j2\pi f_k(\tau-(t-\tau_i-mT_s))}e^{j2\pi\nu_i(\tau+mT_s)}d\tau, \\
&= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{i=1}^{P}\varphi_i x[m,n]e^{j2\pi\nu_i mT_s}e^{j2\pi f_k(t-\tau_i-mT_s)} \\
&\int p^*(\tau-(t-\tau_i-mT_s))g\left(\tau\right)e^{-j2\pi(f_k-\nu_i-nF_0)\tau}d\tau, \\
&= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{i=1}^{P}\varphi_i x[m,n]e^{j2\pi\nu_i mT_s}e^{j2\pi f_k(t-\tau_i-mT_s)} \\
&e^{-j2\pi(f_k-\nu_i-nF_0)(t-\tau_i-mT_s)}\int p^*(\tau-(t-\tau_i-mT_s))g\left(\tau\right) \\
&e^{-j2\pi(f_k-\nu_i-nF_0)(\tau-(t-\tau_i-mT_s))}d\tau, \\
&= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{i=1}^{P}\varphi_i x[m,n]e^{j2\pi\nu_i mT_s}e^{j2\pi(\nu_i+nF_0)(t-\tau_i-mT_s)} \\
&A_{pg}(t-\tau_i-mT_s, f_k-\nu_i-nF_0), \\
&= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{i=1}^{P}\varphi_i x[m,n]e^{-j2\pi\frac{nm}{NM}}e^{j2\pi(\nu_i+nF_0)(t-\tau_i)} \\
&A_{pg}(t-\tau_i-mT_s, f_k-\nu_i-nF_0).
\end{aligned} \tag{29}$$

Combining (29) with the definition of $h_{m,n}(t,f)$ in (8) yields (6).

Appendix B
Proof of Proposition III.1

Since $T_a \ll T$, the time domain supports of $g(t)$ and $p(t)$ are $[0, NT]$ and $[-T, (N+1)T]$, respectively. The cross-ambiguity function between $p(t)$ and $g(t)$ can then be calculated as

$$\begin{aligned}
A_{pg}(t,f) &= \int_{-\infty}^{\infty} p^*(\tau'-t)g(\tau')e^{-j2\pi f(\tau'-t)}d\tau', \\
&= \int_{-\infty}^{\infty} p^*(t')g(t'+t)e^{-j2\pi ft'}dt', \\
&= \sum_{n=-1}^{N}\int_{nT}^{(n+1)T} p^*(t')g(t'+t)e^{-j2\pi ft'}dt', \\
&= \sum_{n=-1}^{N}\int_{0}^{T} a^*(\tau)g(\tau+t+nT)e^{-j2\pi f(\tau+nT)}d\tau,
\end{aligned}$$

$$= \sum_{n=-1}^{N} \sum_{k=0}^{N-1} \int_0^T a^*(\tau) a(\tau - (k-n)T + t) e^{-j2\pi f(\tau+nT)} d\tau,$$

$$= \sum_{k=0}^{N-1} \sum_{n=-1}^{N} A_a(t-(k-n)T, f) e^{-j2\pi nTf},$$

$$= \sum_{k=0}^{N-1} e^{-j2\pi fkT} \times \sum_{n=-1}^{N} A_a(t-(k-n)T, f) e^{j2\pi f(k-n)T}. \quad (30)$$

Since $a(t)$ has a finite time-domain support between $[0, T_a]$, it is straightforward that $A_a(t, f)$ has a finite time domain support between $[-T_a, T_a]$, that is, $A_a(t, f) = 0$ for $|t| > T_a$. The time domain support for $A_a(t-(k-n)T, f)$ is thus

$$(k-n)T - T_a \leq t \leq (k-n)T + T_a. \quad (31)$$

For any given $k \in [0, N-1]$, there is always an $n \in [-1, N]$ such that $k - n = -1, 0$, or 1. Thus for the support $t \in [uT - T_a, uT + T_a]$ with $u = -1, 0$ or 1, we have

$$A_{pg}(t, f) = \sum_{k=0}^{N-1} e^{-j2\pi kTf} \cdot A_a(t-uT, f) e^{j2\pi uTf}, \quad t \in uT + [-T_a, T_a], \quad u = -1, 0, 1 \quad (32)$$

It should be noted that when $|u| > 1$, the relationship $k-n = u$ no longer holds for every $k = [0, N-1]$ due to the finite range of n, thus the relationship in (32) is not applicable to the case $|u| > 1$. This completes the proof.

Appendix C
Proof of Corollary III.1

The support of $A_a(t, f)$ defined in (11) contains three pieces. Based on (11), we will consider the support of $h_{m,n}[l, k]$ by identifying the range of $u = l - m$. Since $0 \leq l, m \leq M-1$, we have $u = l - m \in [-M+1, M-1]$.

1) $|uT_s - (\tau_i - \tau_e)| \leq T_a$, for $i \in \mathcal{P} = \{1, \cdots, P\}$. The range of u can be calculated as

$$u \in \bigcup_{i \in \mathcal{P}} \left[\frac{-T_a + (\tau_i - \tau_e)}{T_s}, \frac{T_a + (\tau_i - \tau_e)}{T_s} \right]. \quad (33)$$

Based on the assumption $|\tau_e| < T_s$, the range of u should be

$$-Q - 1 \leq u \leq Q + 1 + \lfloor \frac{\tau_{\max}}{T_s} \rfloor, \quad (34)$$

or equivalently

$$-L_1 \leq u \leq L_2. \quad (35)$$

2) $T - T_a \leq uT_s - (\tau_i - \tau_e) < T + T_a$, for $i \in \mathcal{P}$. In addition, based on the fact that $u \leq M-1$, the range of u in this case is

$$M - L_1 \leq u \leq M - 1. \quad (36)$$

3) $-T - T_a < uT_s - (\tau_i - \tau_e) \leq -T + T_a$, for $i \in \mathcal{P}$. In this case, based on the fact that $u \geq -M+1$, the range of u is

$$-M + 1 \leq u \leq -M + L_2. \quad (37)$$

The delay domain support for $h_{m,n}[l, k]$ can then be obtained by combining (34) and (37).

Appendix D
Proof of Corollary III.2

When $f = nF_0$, we have $\sum_{k=0}^{N-1} e^{-j2\pi \frac{nk}{N}} = N\delta[n]$. Thus from (12),

$$A_{pg}(mT_s, nF_0) = N\delta[n] \Big[A_a((m-M)T_s, 0) e^{-j2\pi \frac{nM}{N}} + A_a(mT_s, 0) + A_a((m+M)T_s, 0) e^{j2\pi \frac{nM}{N}} \Big]. \quad (38)$$

Since $a(t)$ is a square-root Nyquist filter, we have $A_a(mT_s, 0) = \frac{1}{N}\delta[m]$ and $A_a((m \pm M)T_s, 0) = 0$ for $|m| < M$. This completes the proof.

Appendix E
Proof of Proposition III.2

The elements in the submatrix $\mathbf{H}_{lm}$ are $h_{mn}[l, k]$, for $k, n = 0, \cdots, N-1$. They all share the same delay, $(l-m)T_s$, but with different Doppler spreads, $(k-n)F_0$.

Since $h_{m,n}[l, k]$ has finite support in the delay domain as in Corollary III.1, only a subset of the sub-matrices in $\mathbf{H}$ are non-zero. The indices of the all-zero sub-matrices in $\mathbf{H}$ are those outside $\mathcal{L}$ defined in (14). This completes the proof.

References

[1] Y. Pan, J. Wimer, J. Wu, H. Lin, and J. Yuan, "Cp-free oddm: Modeling and design," in *IEEE International Conference on Communications (ICC) (submitted)*, 2026.

[2] R. Hadani, S. Rakib, M. Tsatsanis, A. Monk, A. J. Goldsmith, A. F. Molisch, and R. Calderbank, "Orthogonal time frequency space modulation," in *2017 IEEE Wireless Communications and Networking Conference (WCNC)*. IEEE, 2017, pp. 1–6.

[3] H. Lin and J. Yuan, "Orthogonal delay-doppler division multiplexing modulation," *IEEE Transactions on Wireless Communications*, vol. 21, no. 12, pp. 11 024–11 037, 2022.

[4] J. Wu and Y. R. Zheng, "Oversampled orthogonal frequency division multiplexing in doubly selective fading channels," *IEEE Transactions on Communications*, vol. 59, no. 3, pp. 815–822, 2010.

[5] J. Wu and P. Fan, "A survey on high mobility wireless communications: Challenges, opportunities and solutions," *IEEE Access*, vol. 4, pp. 450–476, 2016.

[6] W. Zhou, J. Wu, and P. Fan, "High mobility wireless communications with doppler diversity: Fundamental performance limits," *IEEE Transactions on Wireless communications*, vol. 14, no. 12, pp. 6981–6992, 2015.

[7] N. Sun and J. Wu, "Optimum designs for high mobility systems with channel estimation errors," in *2014 IEEE International Conference on Communications (ICC)*. IEEE, 2014, pp. 5072–5077.

[8] L. Gaudio, G. Colavolpe, and G. Caire, "OTFS vs. OFDM in the presence of sparsity: A fair comparison," *IEEE Transactions on Wireless Communications*, vol. 21, no. 6, pp. 4410–4423, 2021.

[9] P. Raviteja, K. T. Phan, Y. Hong, and E. Viterbo, "Interference cancellation and iterative detection for orthogonal time frequency space modulation," *IEEE transactions on wireless communications*, vol. 17, no. 10, pp. 6501–6515, 2018.

[10] C. Shen, J. Yuan, and H. Lin, "Error performance of rectangular pulse-shaped OTFS with practical receivers," *IEEE Wireless Communications Letters*, 2022.

[11] P. Raviteja, Y. Hong, E. Viterbo, and E. Biglieri, "Practical pulse-shaping waveforms for reduced-cyclic-prefix OTFS," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 1, pp. 957–961, 2018.

[12] S. Tiwari and S. S. Das, "Circularly pulse-shaped orthogonal time frequency space modulation," *Electronics Letters*, vol. 56, no. 3, pp. 157–160, 2020.

[13] A. Farhang, A. RezazadehReyhani, L. E. Doyle, and B. Farhang-Boroujeny, "Low complexity modem structure for OFDM-based orthogonal time frequency space modulation," *IEEE Wireless Communications Letters*, vol. 7, no. 3, pp. 344–347, 2017.

[14] S. S. Das, V. Rangamgari, S. Tiwari, and S. C. Mondal, "Time domain channel estimation and equalization of cp-otfs under multiple fractional dopplers and residual synchronization errors," *IEEE Access*, vol. 9, pp. 10 561–10 576, 2020.

[15] Y. Liang, W. Yuan, Y. Wu, and P. Fan, "Matrix pencil based channel estimation algorithm for otfs in fractional doppler channels," *IEEE Communications Letters*, 2024.

[16] H. Liu, Y. Liu, M. Yang, and Q. Zhang, "On the characterizations of otfs modulation over multipath rapid fading channel," *IEEE Transactions on Wireless Communications*, vol. 22, no. 3, pp. 2008–2021, 2022.

[17] Y. Pan, J. Wu, and J. Yuan, "Low complexity otfs detection with a delay-doppler domain cmc-mmse receiver," in *ICC 2024-IEEE International Conference on Communications*. IEEE, 2024, pp. 551–556.

[18] A. Zhou, Y. Pan, J. Wu, H. Lin, and J. Yuan, "On the performance of practical pulse-shaped OTFS with analog receivers," in *IEEE ICC 2023 Workshop on OTFS and Delay-Doppler Multicarrier Communications for 6G*. IEEE, 2023, pp. 1–6.

[19] J. Wexler and S. Raz, "Discrete gabor expansions," *Signal processing*, vol. 21, no. 3, pp. 207–220, 1990.

[20] A. J. Janssen, "Duality and biorthogonality for weyl-heisenberg frames," *Journal of Fourier Analysis and applications*, vol. 1, no. 4, pp. 403–436, 1994.

[21] G. Matz, H. Bolcskei, and F. Hlawatsch, "Time-frequency foundations of communications: Concepts and tools," *IEEE Signal Processing Magazine*, vol. 30, no. 6, pp. 87–96, 2013.

[22] A. Sahin, I. Guvenc, and H. Arslan, "A survey on multicarrier communications: Prototype filters, lattice structures, and implementation aspects," *IEEE communications surveys & tutorials*, vol. 16, no. 3, pp. 1312–1338, 2013.

[23] H. Lin, J. Yuan, W. Yu, J. Wu, and L. Hanzo, "Multi-carrier modulation: An evolution from time-frequency domain to delay-doppler domain," *arXiv preprint arXiv:2308.01802*, 2023.

[24] H. Lin and J. Yuan, "On delay-doppler plane orthogonal pulse," in *GLOBECOM 2022-2022 IEEE Global Communications Conference*. IEEE, 2022, pp. 5589–5594.

[25] K. Huang, M. Qiu, J. Tong, J. Yuan, and H. Lin, "Performance of oddm with imperfect channel estimation," in *2023 IEEE 24th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*. IEEE, 2023, pp. 561–565.

[26] Q. Cheng, Z. Shi, J. Yuan, P. G. Fitzpatrick, and T. Sakurai, "Novel oddm signal detection using contrastive learning for high reliability and fast convergence," in *ICC 2023-IEEE International Conference on Communications*. IEEE, 2023, pp. 1280–1285.

[27] ——, "A novel environmentally robust oddm detection approach using contrastive learning," *IEEE Transactions on Communications*, vol. 71, no. 9, pp. 5274–5286, 2023.

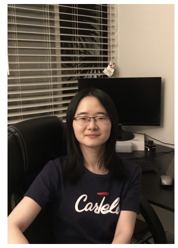
[28] J. Tong, J. Xi, J. Yuan, and H. Lin, "On the input-output relation of oddm modulation over general physical channels," in *2023 IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, 2023, pp. 289–294.

[29] J. Tong, J. Yuan, H. Lin, and J. Xi, "Orthogonal delay-doppler division multiplexing (oddm) over general physical channels," *IEEE Transactions on Communications*, 2024.

[30] ETSI LTE, "Evolved universal terrestrial radio access (E-UTRA); base station (BS) radio transmission and reception (3GPP TS 36.104 version 8.6. 0 release 8), july 2009," *ETSI TS*, vol. 136, no. 104, p. V8, 2009.

[31] C. Wang and Z. Zhou, "A new detection algorithm for ofdm system without cyclic prefix," *Proceedings of the IEEE 6th Circuits and Systems Symposium on Emerging Technologies: Frontiers of Mobile and Wireless Communication (IEEE Cat. No.04EX710)*, p. 453–456, 2004.

[32] J. Wu, Y. R. Zheng, K. B. Letaief, and C. Xiao, "On the error performance of wireless systems with frequency selective fading and receiver timing phase offset," *IEEE Transactions on Wireless Communications*, vol. 6, no. 2, pp. 720–729, 2007.



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